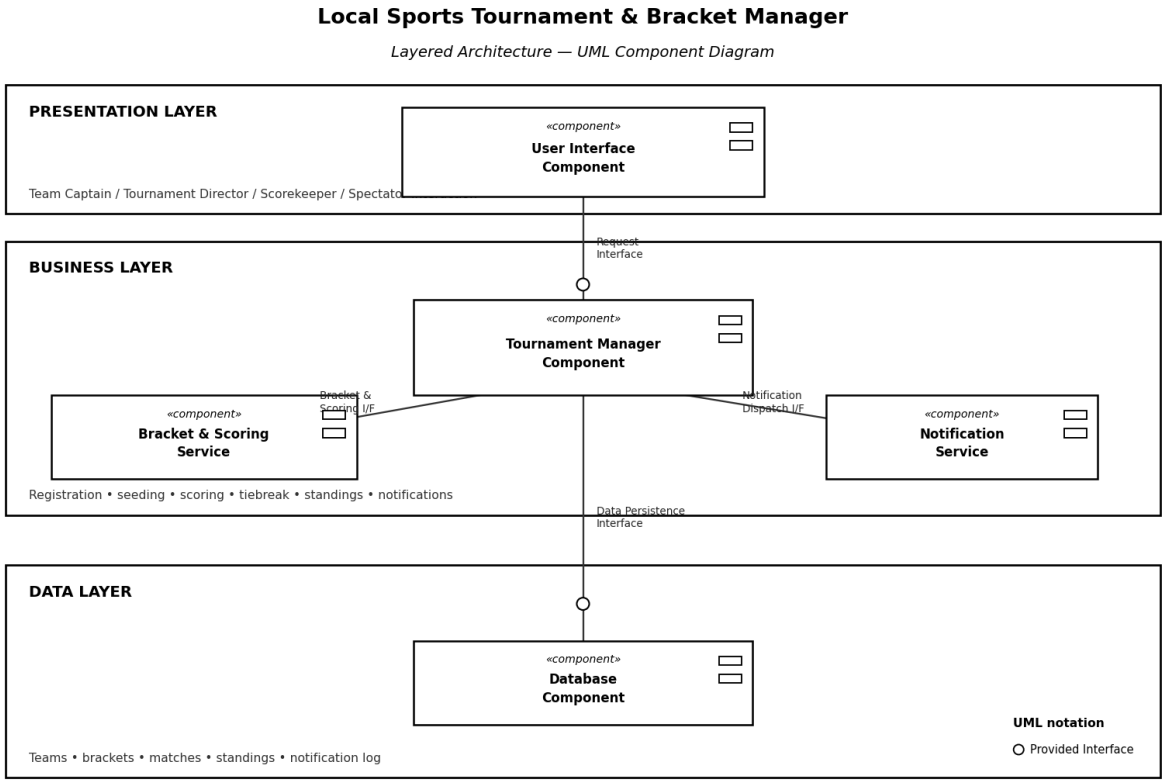


LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Local Sports Tournament & Bracket Manager
SRN: PES1UG24CS914 | Section: 5B | Domain: Media, Events & Community
Users: Team Captain, Tournament Director, Scorekeeper, Spectator

Architecture Selection

We chose **Layered Architecture** for the Local Sports Tournament & Bracket Manager.

Two Reasons

1. Clear separation of responsibilities: The architecture separates presentation, business and data concerns. The User Interface handles interaction for all four actors (Team Captain, Tournament Director, Scorekeeper and Spectator), while the Tournament Manager coordinates the core tournament workflow independently of how each actor's screen is built.

2. Centralized tournament rules: The Tournament Manager coordinates the full workflow of registering teams, generating the seeded bracket, recording match results, applying tiebreaker rules, and publishing standings. Keeping this logic in one place ensures the same rules (e.g. only a Scorekeeper can submit a result, ties are always resolved the same way) are enforced consistently across every use case rather than being duplicated across screens.

Security Advantage

The business layer provides a controlled boundary between the actors and the stored tournament data. Every request — a score submission, a bracket regeneration, a standings publish — is validated by the Tournament Manager before match records, brackets or standings are changed. This supports the security requirements from the SRS: role-based access (only a Tournament Director or assigned Scorekeeper can submit a result) and

preventing duplicate or out-of-order score submissions from corrupting the bracket.

Performance Benefit

The architecture keeps the main scoring flow direct: the User Interface communicates with the Tournament Manager, which coordinates the Bracket & Scoring Service, the Notification Service and the Database directly. Avoiding the additional network and operational overhead of a distributed microservices design keeps the recalculation path short, which supports NFR-001's requirement that standings and tiebreakers recalculate within 200 ms of a score submission.

Selected components: User Interface, Tournament Manager, Bracket & Scoring Service, Notification Service, Database.

Required interfaces: Request Interface, Bracket & Scoring Interface, Notification Dispatch Interface, Data Persistence Interface.